Dataset Name: California Housing Data
Dataset link: https://www.kaggle.com/datasets/camnugent/california-housing-prices/data

Data cleaning and preprocessing: For data cleaning rows containing NULL values were removed, outliers were identified using box plots and rows containing the outliers were removed.

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY
...	...	...	...	...	...	...	...	...	...	...
20635	-121.09	39.48	25.0	1665.0	374.0	845.0	330.0	1.5603	78100.0	INLAND
20636	-121.21	39.49	18.0	697.0	150.0	356.0	114.0	2.5568	77100.0	INLAND
20637	-121.22	39.43	17.0	2254.0	485.0	1007.0	433.0	1.7000	92300.0	INLAND
20638	-121.32	39.43	18.0	1860.0	409.0	741.0	349.0	1.8672	84700.0	INLAND
20639	-121.24	39.37	16.0	2785.0	616.0	1387.0	530.0	2.3886	89400.0	INLAND

20640 rows × 10 columns

Fig: Sample rows of the uncleaned dataset.

We also checked the result by applying winsorizing.

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           20640 non-null  float64
 1   latitude            20640 non-null  float64
 2   housing_median_age  20640 non-null  float64
 3   total_rooms         20640 non-null  float64
 4   total_bedrooms      20433 non-null  float64
 5   population          20640 non-null  float64
 6   households          20640 non-null  float64
 7   median_income       20640 non-null  float64
 8   median_house_value  20640 non-null  float64
 9   ocean_proximity     20640 non-null  object
dtypes: float64(9), object(1)
memory usage: 1.6+ MB
None
```

Fig: Dataset information showing Null count, data type and column names.

Columns that are deemed not necessary or dummy to the analysis were removed.

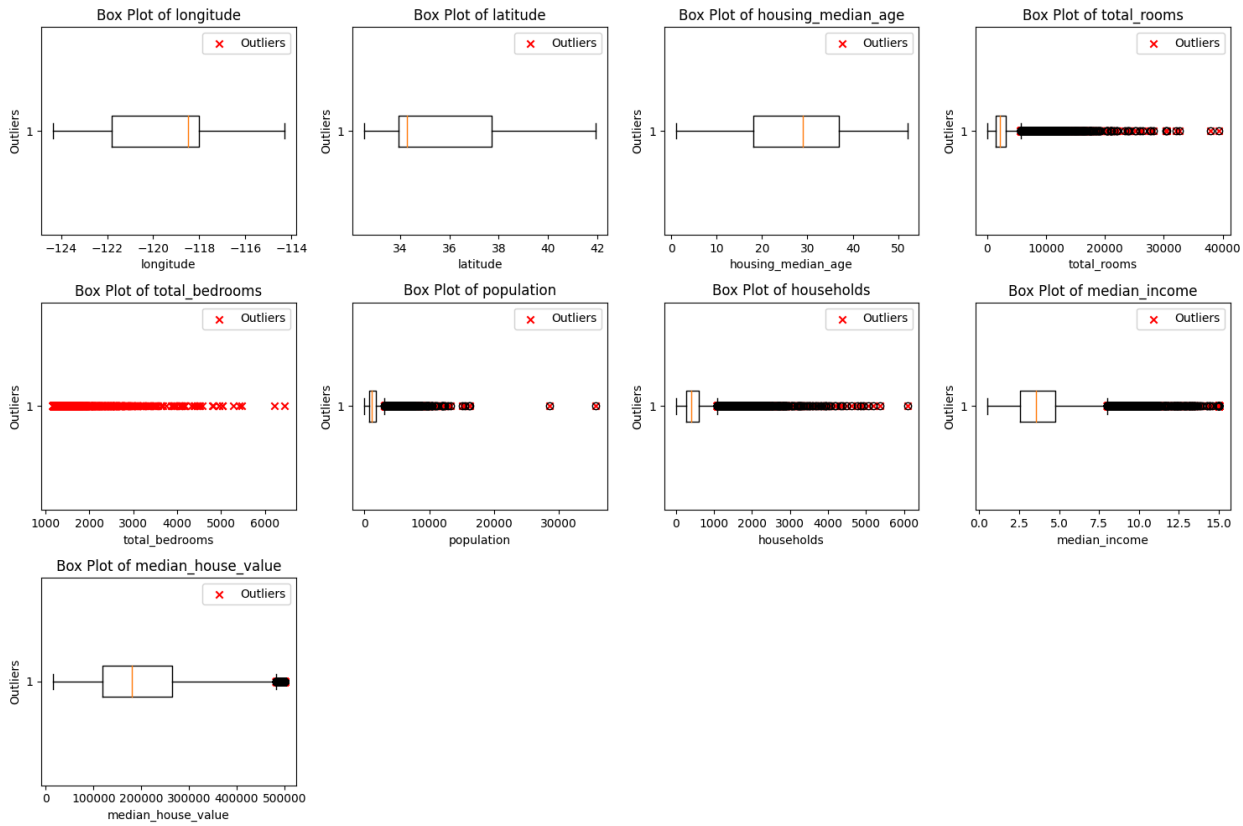

Fig: Box plotting to Visualize the outliers

Correlation between the features were identified using correlation matrix and heat mapping and features with relation above 0.60 were removed.

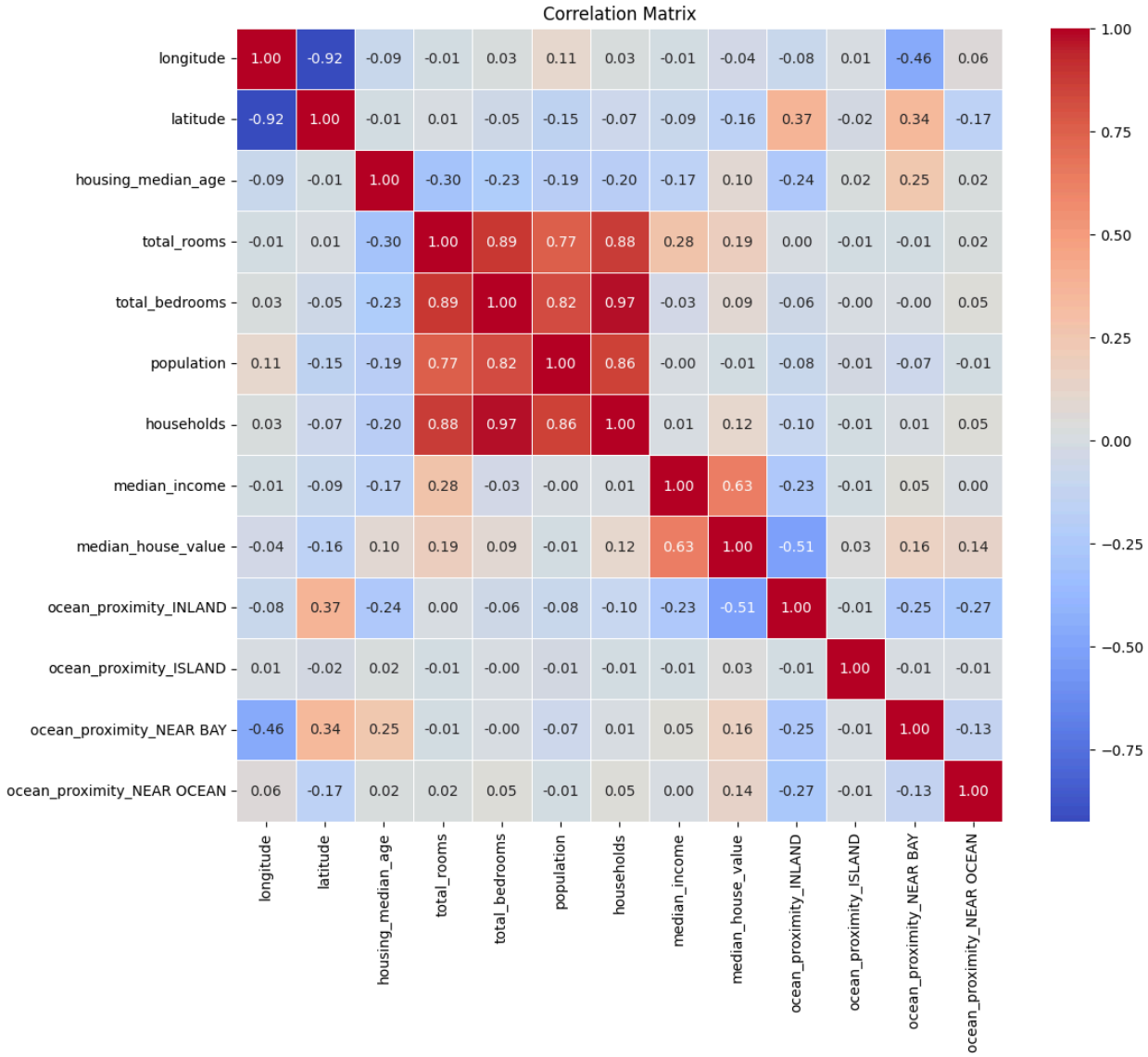

Fig: Heat mapping showing the correlation between all the features.

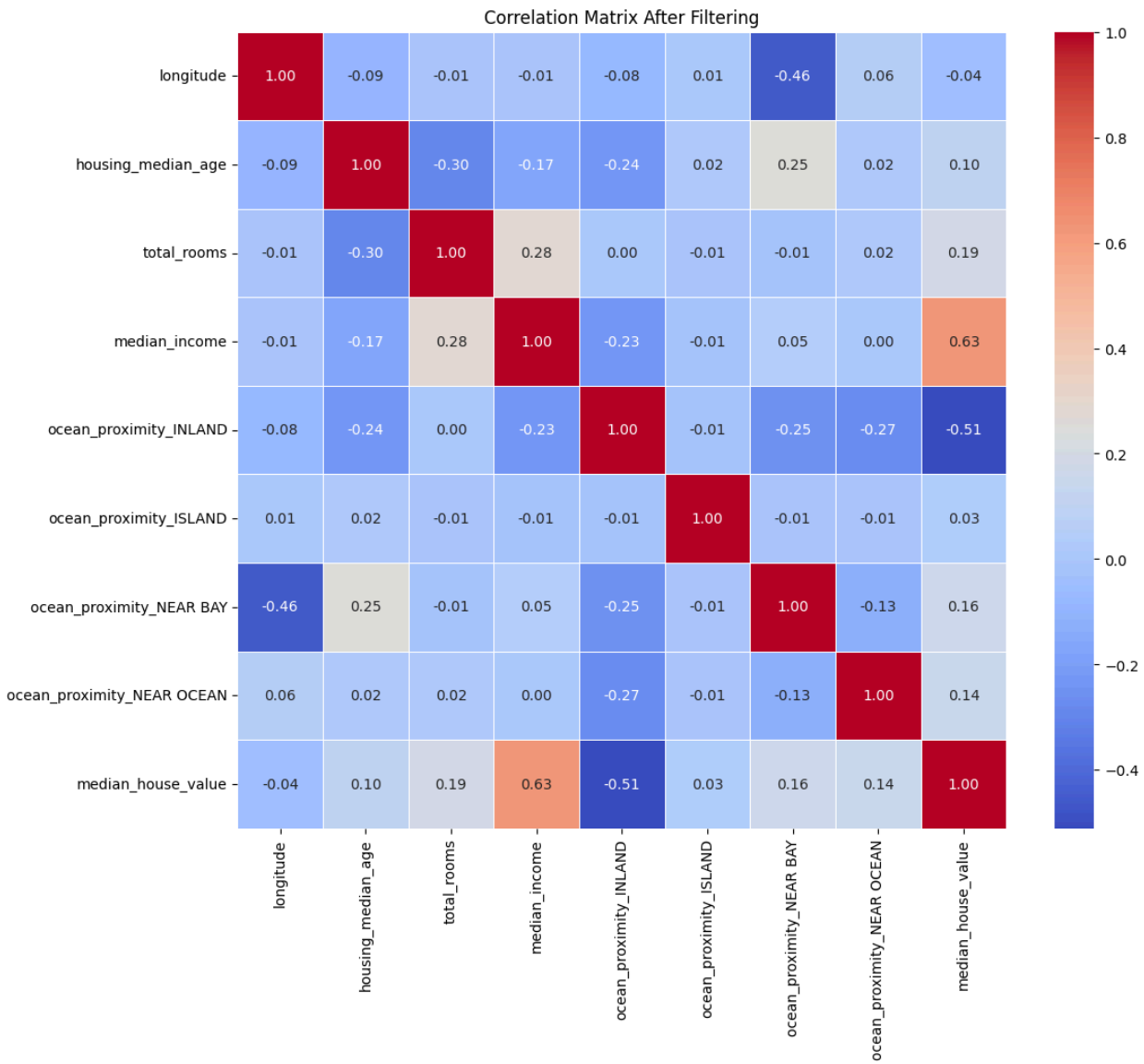

Fig: heat mapping showing correlation between the variables after removing the features with high correlation

The ocean_proximity feature contained categorical values so one-hot encoding was applied to convert it into numeral values.

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity_INLAND	ocean_proximity_ISLAND	ocean_proximity_NEAR BAY	ocean_proximity_NEAR OCEAN
0	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	0	0	1	0
1	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	0	0	1	0
2	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	0	0	1	0
3	-122.25	37.85	52.0	919.0	213.0	413.0	193.0	4.0368	269700.0	0	0	1	0
4	-122.25	37.84	52.0	2535.0	489.0	1094.0	514.0	3.6591	299200.0	0	0	1	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
17616	-121.09	39.48	25.0	1665.0	374.0	845.0	330.0	1.5603	78100.0	1	0	0	0
17617	-121.21	39.49	18.0	697.0	150.0	356.0	114.0	2.5568	77100.0	1	0	0	0
17618	-121.22	39.43	17.0	2254.0	485.0	1007.0	433.0	1.7000	92300.0	1	0	0	0
17619	-121.32	39.43	18.0	1860.0	409.0	741.0	349.0	1.8672	84700.0	1	0	0	0
17620	-121.24	39.37	16.0	2785.0	616.0	1387.0	530.0	2.3886	89400.0	1	0	0	0

17621 rows × 13 columns

Fig: sample rows of the cleaned dataset.

Model training and analysis: Linear regression model was trained using both Python and R programming environments. After training, r-squared 0.62-0.66 was achieved on all possible) combinations(more than 4000 of the features. Therefore the linear regression model is capable of explaining about 62-66% of the dataset.

```
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -2.239e+06  7.974e+04 -28.082  < 2e-16 ***
longitude           -2.636e+04  9.335e+02 -28.233  < 2e-16 ***
latitude            -2.579e+04  9.354e+02 -27.573  < 2e-16 ***
housing_median_age   1.215e+03  4.159e+01  29.209  < 2e-16 ***
OceanProximity_Inland -3.323e+04  1.639e+03 -20.272  < 2e-16 ***
OceanProximity_Island  1.363e+05  2.925e+04   4.659  3.2e-06 ***
median_income_Win    4.754e+04  4.167e+02 114.096  < 2e-16 ***
total_rooms_Win     -1.947e+01  1.207e+00 -16.125  < 2e-16 ***
total_bedrooms_Win   2.678e+02  5.833e+00  45.917  < 2e-16 ***
population_Win      -5.941e+01  1.276e+00 -46.558  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 65350 on 20423 degrees of freedom
Multiple R-squared:  0.6672,    Adjusted R-squared:  0.667
F-statistic:  4549 on 9 and 20423 DF,  p-value: < 2.2e-16
```